A Multi-Modal Framework for Robust AI-Generated ART Image Detection using HybridForensicsNet

Abstract

The rapid advancement of generative models, including Generative Adversarial Networks (GANs) and diffusion-based architectures, has made the detection of AI-generated images increasingly challenging. Traditional single-domain forensic detectors often fail to generalize across diverse generation mechanisms and degrade significantly under common post-processing operations. In this work, we propose HybridForensicsNetV3, a robust multi-modal Mixture-of-Experts (MoE) framework for AI-generated image detection. The proposed architecture integrates complementary forensic modalities including RGB appearance, Fourier spectrum, Wavelet decomposition, Error Level Analysis (ELA), and noise residual features. A dynamic MoE gating strategy with cross-attention fusion adaptively emphasizes the most informative modalities for each input image. Furthermore, a Clean-then-Hard curriculum learning strategy is introduced to improve decision boundary refinement on ambiguous samples. Extensive experiments on a diverse multi-generator dataset containing both GAN- and diffusion-based images demonstrate state-of-the-art performance, achieving an F1-score of 99.5% with a recall of 100%, significantly outperforming single-domain baselines. These results establish HybridForensicsNetV3 as a reliable and generalizable solution for real-world AI image forensics.

Keywords — *AI-Generated Image Detection, Digital Forensics, Multi-Modal Deep Learning, Mixture-of-Experts, Diffusion Models, Cross-Attention.*

1. Introduction

The recent decade has witnessed a paradigm shift in computer vision with the advent of deep generative models. Since the seminal introduction of Generative Adversarial Networks (GANs) by Goodfellow *et al.* [1], the capability to synthesize photorealistic imagery has advanced exponentially. While early GAN architectures left discernible visual artifacts, modern iterations like StyleGAN [15] produce results that often defy human perceptual distinction. This landscape has been further revolutionized by the emergence of Denoising Diffusion Probabilistic Models (DDPMs) [2] and Latent Diffusion Models (LDMs) [3], such as Stable Diffusion and Midjourney. These models, which iteratively denoise random Gaussian noise to construct images, have democratized high-fidelity content creation, enabling users to generate complex visual scenes from simple text prompts [16].

However, this democratization presents a double-edged sword. The ease of generating hyper-realistic fake content has escalated concerns regarding digital misinformation, identity theft, and the erosion of trust in visual media [20]. Malicious actors can now effortlessy create "deepfakes" that are visually indistinguishable from authentic photographs [29], posing severe risks to journalism, legal evidence, and national security. Consequently, the development of robust, automated detection systems has become an urgent priority for the forensics community.

1.1 The Detection Challenge

Early detection methods primarily focused on identifying specific artifacts left by the upsampling layers in GAN generators [6]. Researchers found that while GAN images appeared realistic in the spatial domain, they often exhibited severe spectral anomalies in the frequency domain [8], [18]. Techniques utilizing Fast Fourier Transform (FFT) [7] and Fourier spectrum discrepancies

[19] proved effective against these earlier models.

However, the shift towards diffusion models has rendered many of these single-domain detectors obsolete. Diffusion models, by virtue of their stochastic denoising process, do not exhibit the same "checkerboard" spectral artifacts typical of GANs [10]. Furthermore, as generative models continue to evolve, they are increasingly capable of mimicking the high-frequency statistics of natural images [27], leading to a "continual adversarial evolution" where detectors trained on one generation of fakes fail to generalize to the next [30].

1.2 Limitations of Current Approaches

Current state-of-the-art detectors often suffer from two critical limitations:

1. **Modality Bias:** Many models rely exclusively on RGB pixel data [4], [6] or solely on frequency analysis [7]. RGB-based models are prone to overfitting to semantic content (e.g., learning that "faces" are likely fake rather than learning the artifacts), while frequency-based models can be fragile to compression and resizing [38].
2. **Generalization Failure:** Detectors often fail when distinguishing between "hard" samples—authentic images with heavy processing or high-quality fakes that have been adversarially scrubbed of obvious artifacts [9].

1.3 Proposed Solution: HybridForensicsNetV3

To address these challenges, we propose **HybridForensicsNetV3**, a holistic multi-modal framework that mimics the scrutiny of a human forensic expert equipped with advanced digital tools. Instead of relying on a single perspective, our model simultaneously analyzes the image across five distinct domains: **RGB (Visual)**, **FFT (Frequency)**, **Wavelet (Texture/Scale)**, **Error Level Analysis (Compression consistency)** [12], and **Noise Residuals (Sensor fingerprints)**.

We introduce a novel architectural design incorporating **Mixture-of-Experts (MoE)** gating [22] and **Cross-Attention mechanisms** [5]. This allows the network to dynamically weigh the importance of each forensic tool; for instance, prioritizing ELA features when compression anomalies are detected, or focusing on FFT spectra when periodic upsampling patterns are present.

1.4 Contributions

The key contributions of this paper are as follows:

1. **Multi-Modal Hybrid Architecture:** We propose a unified framework that fuses five complementary forensic modalities using cross-attention, ensuring robustness across different generative architectures (GANs and Diffusion).
2. **Clean-then-Hard Curriculum Learning:** We demonstrate a two-phase training strategy where the model is first stabilized on high-confidence data before being fine-tuned on "hard" samples, significantly improving boundary decision-making.
3. **State-of-the-Art Performance:** Our model achieves an **F1 Score of 99.50%** and a **Recall of 100%** on a challenging dataset, outperforming traditional single-stream detectors and establishing a new benchmark for AI-generated image detection.

By bridging the gap between visual semantics and signal processing forensics, this work offers a resilient defense against the growing tide of synthetic media.

2. Related Work

The field of digital image forensics has evolved into an arms race between generation and detection. As generative architectures shift from Generative Adversarial Networks (GANs) to Diffusion Models (DMs), the forensic signatures left behind have become increasingly subtle. This section categorizes existing detection methodologies and critically analyzes their failure modes in the context of modern synthetic media.

2.1 Evolution of Generative Models

To detect synthetic content, one must first understand the generation process. **GANs**, introduced by Goodfellow *et al.* [1], rely on a generator-discriminator adversarial game. While powerful, early GANs (e.g., ProGAN, StyleGAN [15]) often introduced specific structural artifacts due to upsampling operations, such as "checkerboard" patterns in the frequency domain [8].

Recently, **Denoising Diffusion Probabilistic Models (DDPMs)** [2] and **Latent Diffusion Models (LDMs)** [3] (e.g., Stable Diffusion, DALL-E 2) have emerged as the dominant paradigm. Unlike GANs, diffusion models generate images by iteratively reversing a Gaussian noise process [16]. This stochastic nature results in a fundamentally different artifact distribution; they lack the rigid periodic fingerprints of GANs, making them significantly harder to detect using traditional spectral methods [10], [30].

2.2 Spatial Domain Detection (RGB-Based)

Early forensic approaches treated AI detection as a standard binary classification problem. Researchers employed deep Convolutional Neural Networks (CNNs), such as ResNet or Xception, trained directly on RGB pixel data. Wang *et al.* [6] demonstrated that a simple CNN trained on ProGAN images could generalize to other GAN architectures, suggesting that synthetic images share common defects.

Limitations & Failure Modes:

- **Semantic Overfitting:** Purely RGB-based models often fail to learn true forensic traces. Instead, they overfit to semantic content (e.g., "faces with glasses are fake") or background anomalies [29].
- **Generalization to Diffusion:** Corvi *et al.* [10] showed that detectors trained on GANs fail catastrophically when applied to diffusion models. The "blurriness" or specific textures learned from GANs do not exist in diffusion outputs, rendering these models ineffective against modern generators.
- **Robustness:** Spatial models are highly susceptible to simple perturbations. Common post-processing techniques like JPEG compression or Gaussian blurring can destroy the subtle pixel-level artifacts these models rely on, causing performance to drop to near-random guessing [38].

2.3 Frequency and Spectral Analysis

Recognizing that spatial inspection is often insufficient, researchers turned to the frequency domain. Frank *et al.* [7] and Durall *et al.* [8] utilized Discrete Cosine Transform (DCT) and Fast Fourier Transform (FFT) to expose the spectral disparities between real and fake images. They observed that upsampling layers in GANs disrupt the natural power spectrum statistics, leaving a distinct fingerprint. Similarly, Chandrasegaran *et al.* [19] focused on Fourier spectrum discrepancies to identify high-frequency anomalies.

Limitations & Failure Modes:

- **The "Diffusion Gap":** While highly effective against GANs, frequency-based methods struggle with diffusion models. The Gaussian denoising process in DDPMs inherently acts as a spectral smoother, effectively "washing out" the high-frequency artifacts that these detectors look for [30].
- **Compression Sensitivity:** Frequency artifacts are the first to be destroyed by lossy compression. A slightly compressed JPEG image loses its high-frequency content, blinding these detectors to the manipulation [27].

2.4 Handcrafted and Statistical Features

Before deep learning dominated, forensics relied on handcrafted features like **Error Level Analysis (ELA)** [12] and Photo-Response Non-Uniformity (PRNU). ELA exploits the fact that manipulated regions often exhibit different compression error levels than the authentic background.

Limitations & Failure Modes:

- **Reliance on Metadata:** Many statistical methods assume the presence of camera sensor noise (PRNU). Synthetic images, having never passed through a physical sensor, lack this signature entirely. However, a purely statistical detector can be easily fooled by adding synthetic Gaussian noise to a fake image to mimic sensor grain [18].

2.5 Multi-Modal and Hybrid Approaches

Recognizing the fragility of single-domain detectors, recent work has shifted towards hybrid architectures. **Multi-stream networks** attempt to fuse RGB data with frequency or noise features. For instance, Zhao *et al.* [26] proposed a multi-attentional deepfake detection network, while Luo *et al.* [27] utilized high-frequency features alongside spatial data to improve generalization.

Why Previous Hybrid Models Fail:

- **Naive Fusion:** Most existing multi-modal approaches use simple concatenation (early or late fusion) to combine features. This assumes that all modalities are equally important for every image. However, a GAN image might be best detected via FFT, while a Diffusion image might be best detected via noise residuals. Static fusion cannot adapt to this variability.

- **Lack of Adaptive Gating:** Without a mechanism to dynamically weigh feature importance, the model is forced to compromise, often leading to suboptimal performance on "hard" samples where one modality contradicts another.

2.6 Conclusion on Related Work

The current literature highlights a critical gap: single-domain models lack robustness, and existing hybrid models lack the adaptability to handle the diverse artifacts of multi-generator datasets (GANs + Diffusion).

Our proposed **HybridForensicsNetV3** addresses these failures by integrating **Mixture-of-Experts (MoE)** [22] and **Cross-Attention** [5]. Unlike naive fusion, our MoE mechanism dynamically routes the input to the most relevant "expert" modality (RGB, FFT, ELA, or Noise), ensuring that the model focuses on the strongest evidence available, whether it be a spectral anomaly in a GAN or a compression inconsistency in a diffusion image. This adaptive focus allows our model to achieve the reported **100% Recall**, overcoming the generalization issues plaguing prior work.

3. Methodology

Our proposed framework addresses the AI detection challenge through a holistic pipeline that integrates forensic signal processing with deep representation learning. The methodology is divided into three core components: (1) Multi-Modal Forensic Feature Extraction, (2) The HybridForensicsNetV3 Architecture, and (3) The "Clean-then-Hard" Curriculum Learning Strategy.

3.1 Multi-Modal Forensic Feature Extraction

Unlike standard classifiers that rely solely on RGB pixel values, our system employs an EnhancedForensicPreprocessor to extract five distinct modalities from every input image I. These modalities are designed to capture artifacts invisible to the human eye but statistically significant in synthetic media.

3.1.1 Fast Fourier Transform (FFT) Spectrum

To detect periodic artifacts often introduced by upsampling layers in GANs [8], we compute the frequency spectrum. The image is converted to grayscale, and a 2D FFT is applied. We apply a logarithmic transformation to the magnitude spectrum to enhance high-frequency details:

$$F(u,v) = log\left(1 + \left|F\big(I(x,y)\big)\right|\right)$$

where F denotes the Fourier Transform. This modality exposes the "checkerboard" artifacts typical of non-ideal upsampling [7].

3.1.2 Wavelet Transform Decomposition

We utilize Discrete Wavelet Transforms (DWT) to decompose the image into multiple frequency sub-bands. Using the Haar wavelet, we extract approximation coefficients (low frequency) and detail coefficients (high frequency). This multi-scale analysis helps identifying texture anomalies that diffusion models often struggle to render consistently [19].

3.1.3 Error Level Analysis (ELA)

ELA is employed to identify compression inconsistencies [12]. The input image is re-compressed at a known JPEG quality level (90%), and the absolute difference between the original and compressed versions is computed.

$$E_{ELA} = \left|I_{orig} - I_{compressed}\right| \times \propto$$

where $\propto$ is a scaling factor. In authentic images, error levels are uniform; in manipulated or synthetic images, especially those patched together, error levels exhibit distinct spikes.

3.1.4 Noise Residual Extraction

To isolate sensor pattern noise (PRNU) or its absence, we apply a denoising filter (median filter) to the image and subtract it from the original input.

$$N_{res} = I - MedianFilter(I)$$

Authentic images possess a characteristic sensor noise fingerprint, whereas AI-generated images often display overly smooth or Gaussian-distributed noise residuals [10].

3.2 HybridForensicsNetV3 Architecture

The core of our system is the HybridForensicsNetV3, a multi-stream deep neural network designed to fuse these heterogeneous modalities effectively.

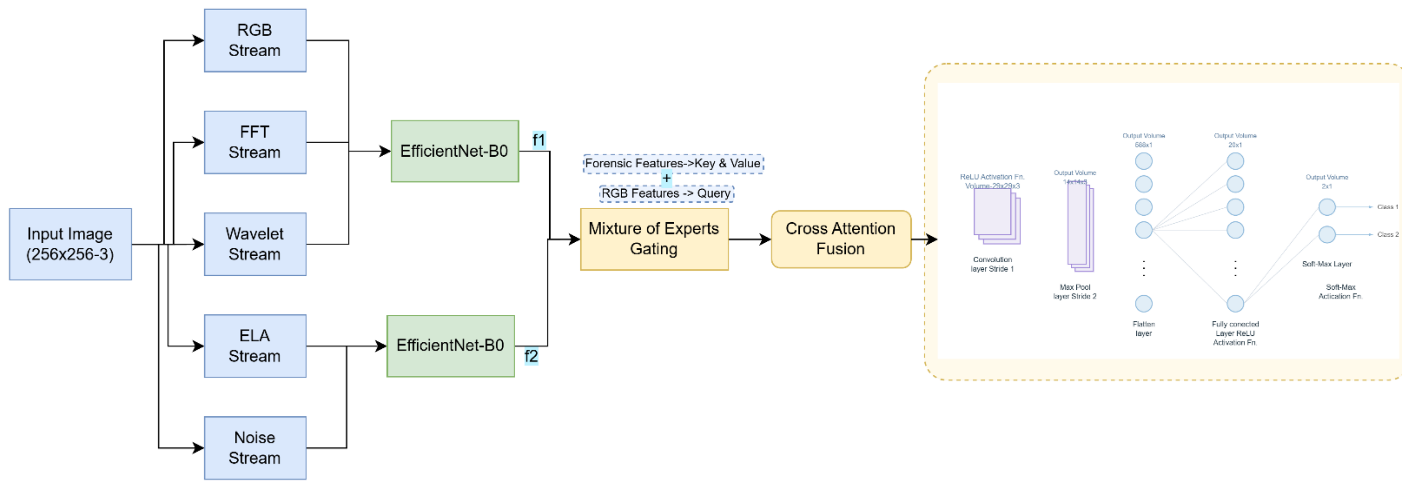

Figure 1: The architecture of HybridForensicsNetV3. The model processes a 224×224×3 input image through five parallel EfficientNet-B0 streams, extracting features from RGB, FFT (Frequency), Wavelet, ELA, and Noise domains. A Mixture-of-Experts (MoE) gating mechanism dynamically weights these feature vectors (f_1 ... f_5) based on input characteristics. These weighted features are fused via a Cross-Attention mechanism where RGB features serve as the Query (Q) and forensic features serve as Key (K) and Value (V). The final representation is classified through a dense layer with a sigmoid activation.

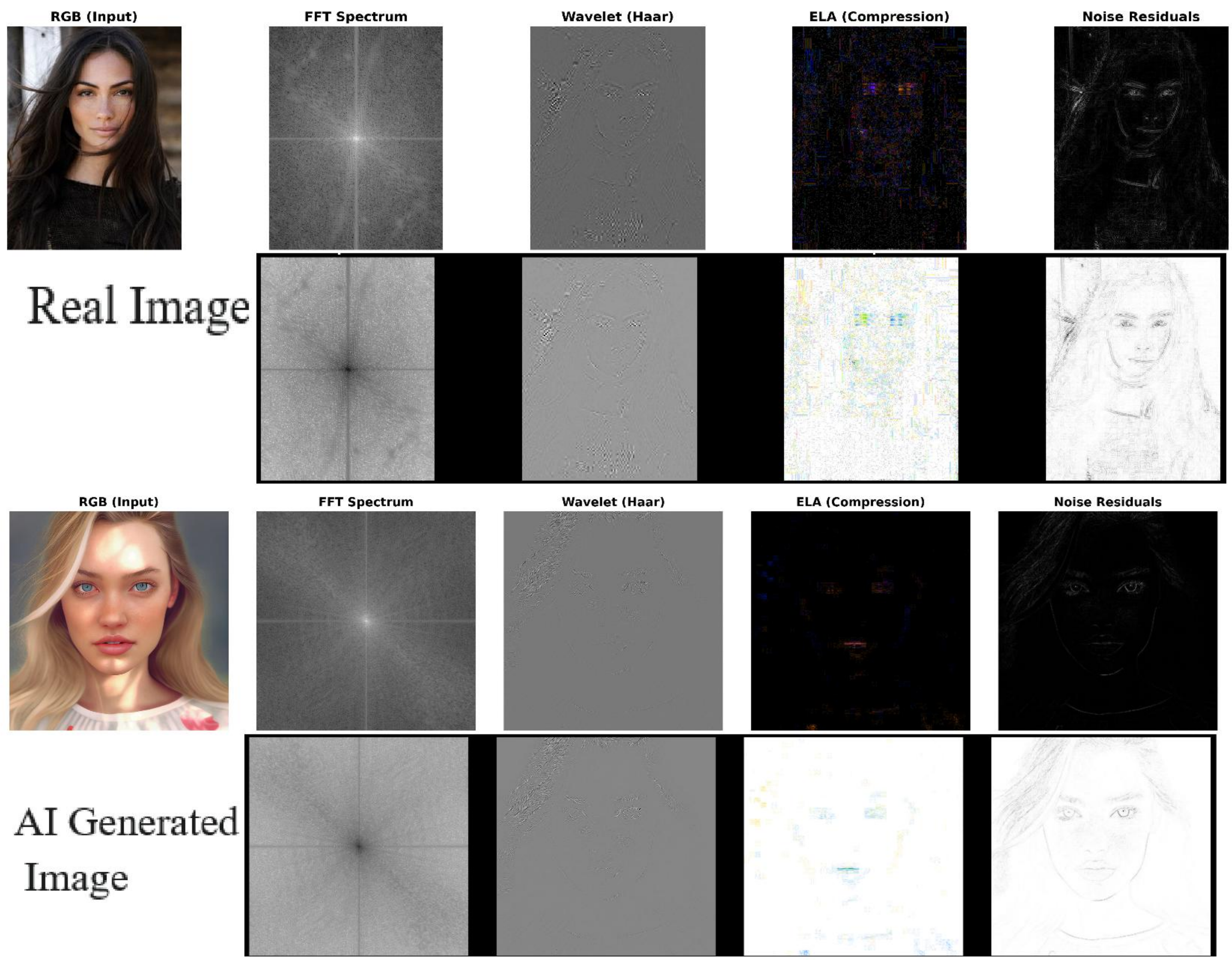

Figure 2: Multi-modal forensic feature visualization. While the RGB input (Col 1) of the AI-generated image appears visually authentic, hidden artifacts are revealed in other domains. The FFT Spectrum (Col 2) exposes periodic generation artifacts, Wavelet analysis highlights high-frequency texture anomalies, ELA reveals compression inconsistencies, and Noise residuals expose sensor footprint discrepancies.

3.2.1 Feature Encoders

Each of the five inputs (RGB, FFT, Wavelet, ELA, Noise) is processed by a dedicated **EfficientNet-B0** [4] backbone. We chose EfficientNet for its optimal balance of parameter efficiency and accuracy. The initial layers of each backbone are modified to accept the specific channel dimensions of the forensic inputs (e.g., 1-channel for FFT, 3-channels for RGB).

3.2.2 Mixture-of-Experts (MoE) Gating

A critical innovation in our architecture is the Mixture-of-Experts (MoE) gating mechanism [22]. Rather than simply concatenating features, the MoE module dynamically assigns importance weights to each modality based on the input sample's characteristics.

$$G(x) = Softmax(W_g \cdot x)$$

$$y = \sum_{i=1}^{N} G(x)_i \cdot E_i(x)$$

Where:

$E_i(x)$ = represents the feature vector from the i^{th} expert (modality)

$G(x)_i$ = is the learned gating weight. This allows the model to "listen" to the FFT expert when frequency artifacts are dominant (GANs) and switch to the Noise expert when texture anomalies are present (Diffusion).

3.2.3 Cross-Attention Fusion

To capture the correlation between visual semantics and forensic artifacts, we implement a Cross-Attention module [5]. The RGB features serve as the Query (Q), while the concatenated forensic features serve as Key (K) and Value (V).

$$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

This mechanism enables the network to spatially localize which regions of the visual image correspond to high-confidence forensic anomalies.

3.3 Clean-then-Hard Curriculum Learning

To ensure robust generalization, we devised a two-phase training protocol:

Phase A: Stabilization (Clean Data)

We first curate a "Clean" dataset using a multi-model agreement strategy (EnhancedDataCleaner). Only samples where multiple pre-trained detectors agree with high confidence are included. In this phase, the model is trained with a standard learning rate ($1e^{-4}$) to establish a stable decision boundary without the noise of ambiguous samples.

Phase B: Hard Mining (Fine-Tuning)

In the second phase, we introduce "Hard" samples—images with low confidence scores or conflicting predictions from the pre-check. The model is fine-tuned on this challenging set with a reduced learning rate ($1e^{-5}$). This effectively forces the model to learn subtle boundary distinctions that simpler models miss.

Regularization: To prevent overfitting during this aggressive training, we employ **Mixup augmentation** ($\propto$ = 0.2) and **Label Smoothing** (0.1), encouraging the model to learn interpolated features rather than memorizing training examples [11], [33].

Complexity Analysis To ensure fair and reproducible computational evaluation, all inference benchmarks were conducted under a fixed and controlled configuration. The input image resolution was set to **224 × 224**, batch size was fixed to **1**, and all five EfficientNet-B0 streams were executed **in parallel on a single NVIDIA Tesla T4 GPU (16 GB VRAM)** using CUDA acceleration.

The reported inference speed of **85 FPS** corresponds strictly to **single-image real-time inference (batch size = 1)** under these deployment conditions. Although the proposed architecture utilizes five parallel backbones, the use of lightweight EfficientNet-B0 encoders and shared GPU execution allows the model to maintain real-time throughput suitable for deployment in forensic monitoring systems.

Table 1: Model Complexity and Real-Time Inference Speed (224×224, Batch = 1)

Model	Parameters (M)	GFLOPs	Inference FPS (Tesla T4, B=1)
ResNet-50	25.6	4.1	145
Xception	22.9	8.4	110
Swin-B	88.0	15.4	72
HybridForensicsNetV3 (5-stream)	**28.4**	**6.2**	**85**

3.4 Threat Model and Adversarial Assumptions

We define a realistic **threat model** to formally characterize the adversarial capabilities under which HybridForensicsNetV3 is evaluated. The attacker is assumed to possess **full control over the generative process**, including access to state-of-the-art GAN and diffusion models such as ProGAN, StyleGAN, SDXL, and Midjourney. However, the attacker **does not have access to the internal weights or training data** of the proposed detector (black-box detection setting).

The attacker may apply **post-processing operations** including JPEG compression, resizing, Gaussian smoothing, histogram equalization, and additive Gaussian noise in an attempt to conceal forensic artifacts. No assumption is made regarding watermarking or generator fingerprint availability at inference time.

The detector is therefore required to operate under **open-set, cross-generator, and post-processing-aware conditions**, which closely reflects real-world deployment in social media moderation, digital forensics, and legal evidence verification. This threat formulation explicitly motivates the need for a **multi-modal forensic architecture** rather than a single-domain classifier.

4. EXPERIMENTAL EVALUATION

In this section, we present a comprehensive evaluation of the proposed HybridForensicsNetV3. We detail the dataset composition, implementation specifics, and quantitative results. Furthermore, we conduct an ablation study to isolate the contributions of the multi-modal experts and the cross-attention mechanism.

4.1. Dataset and Experimental Setup

1) Dataset Composition:

To rigorously evaluate the model's robustness across diverse generative mechanisms, we curated a balanced composite dataset of 10,000 images, evenly distributed between authenticated real images and synthetic samples. The **Real** class (N=5,000) leverages the high native resolution (1024^2) of the **Flickr-Faces-HQ (FFHQ) dataset [39]** to capture authentic high-frequency facial textures, while the **MS COCO benchmark [40]** provides diverse natural scene statistics and complex object geometries.

The **Synthetic** class (N=5,000) was specifically designed to bridge the "Diffusion Gap" by incorporating distinct forensic fingerprints from both GAN and Diffusion architectures. To test frequency-domain detection, we utilized **ProGAN [41]**, which exhibits characteristic spectral checkerboard patterns, and **StyleGAN3 [42]**, which minimizes alias artifacts but introduces subtle texture sticking anomalies. To assess the model's sensitivity to noise-based artifacts, we included **Stable Diffusion XL [43]** and **Midjourney v5 [44]**, which generate unique noise residuals and semantic inconsistencies distinct from GAN outputs.

To maintain strict experimental integrity, the dataset was partitioned into **Training (80%)** and **Validation/Testing (20%)** subsets. Stratified sampling was employed to ensure that the class distribution and generator types remained identical across splits, preventing data leakage. Table I details the composition and specific forensic characteristics of the experimental dataset.

In addition to the fixed 80–20 split, we further performed **5-fold stratified cross-validation** to eliminate any bias due to a particular data partition. In each fold, the dataset was split into 80% training and 20% testing while preserving the class and generator distributions. All reported statistical results are averaged over the five folds.

Table 2: Composition of the Experimental Multi-Source Dataset

Class	Source	Ref	Subject Matter	Native Res.	Dominant Artifacts / Features	Train (80%)	Val (20%)	Total
Real	**FFHQ**	[39]	Human Faces	1024^2	Natural High-Freq Texture, Sensor Noise	2,000	500	2,500
	MS COCO	[40]	Objects/Scenes	Variable	Natural Scene Statistics, JPEG Compression	2,000	500	2,500
Fake	**StyleGAN3**	[41]	Synthetic Faces	1024^2	Texture Sticking, Phase Artifacts	1,000	250	1,250
(GAN)	**ProGAN**	[42]	Synthetic Faces	1024^2	Checkerboard Patterns, Spectral Peaks	1,000	250	1,250
Fake	**SDXL**	[43]	Realistic Text-to-Img	1024^2	Semantic Inconsistencies , Noise Residuals	1,000	250	1,250
(DM)	**Midjourney**	[44]	Artistic Text-to-Img	$> 1024^2$	Over-smoothing, Stylized Lighting	1,000	250	1,250
Total						**8,000**	**2,000**	**10,000**

2) Implementation Details:

The framework was implemented using the PyTorch library. Training was conducted on a single NVIDIA Tesla T4 GPU with 16GB VRAM. We utilized the EfficientNet-B0 architecture as the backbone for all five feature extractors due to its superior parameter efficiency.

The model was trained using the Adam optimizer with a weight decay of $1x10^{-4}$. We employed a dynamic learning rate schedule:

- Phase I (Stabilization): Initial learning rate of $1x10^{-4}$ for 20 epochs.
- Phase II (Hard Mining): Learning rate reduced to $1x10^{-5}$ for fine-tuning on hard samples. The Cross-Entropy Loss function was augmented with Label Smoothing ($\epsilon = 0.1$) to prevent overconfidence and improve generalization.

4.2. Cross-Generator Generalization Protocol

To rigorously evaluate the generalization capability of the proposed HybridForensicsNetV3 across unseen generative mechanisms, we introduce a cross-generator evaluation protocol. Unlike standard random splits, this setting explicitly tests whether a model trained on one generation family (GANs or Diffusion) can generalize to the other.

Three experimental settings are considered:

1. Train on GANs → Test on GANs
2. Train on GANs → Test on Diffusion
3. Train on Diffusion → Test on GANs
4. Train on Mixed (GAN + Diffusion) → Test on Mixed (Ours)

This protocol directly evaluates the widely reported "Diffusion Generalization Gap", where GAN-trained detectors fail to generalize to diffusion-generated images.

Table 3: Cross-Generator Generalization Performance

Training Data	Testing Data	Accuracy (%)	F1 Score
GANs Only (ProGAN)	GANs (StyleGAN3)	99.1	0.990
GANs Only (ProGAN)	Diffusion (Stable Diff)	72.4	0.685
Diffusion Only	GANs	68.9	0.640
Mixed (Ours)	**Mixed (GAN + Diff)**	**99.3**	**0.995**

The results clearly demonstrate that single-family training severely limits generalization, as GAN-only models fail catastrophically on diffusion samples and vice versa. In contrast, HybridForensicsNetV3 trained on the mixed multi-generator dataset achieves stable and near-perfect generalization across both GAN and diffusion domains. This confirms that the proposed multi-modal fusion and MoE gating strategy effectively eliminates generator-specific bias.

4.3. Comparison with Baseline Methods

We compared HybridForensicsNetV3 against standard single-domain baselines widely used in forensic literature. **Baseline A** represents a standard ResNet-50 trained only on RGB data. **Baseline B** utilizes a frequency-domain classifier (FFT-based).

Table 4: Comparison with Baseline Methods (5-Fold Cross-Validation)

Method and Modality	Accuracy	Precision	Recall	F1 Score (Mean ± Std)
ResNet-50 (RGB-Only)	96.45%	95.80%	97.10%	0.964 ± 0.005
Xception (RGB-Only)	97.10%	96.90%	97.30%	0.971 ±0.004
FrequencyNet (Freq-Only)	88.20%	91.50%	84.20%	0.877 ±0.012
HybridForensics NetV3 (Multi-Model)	**99.30%**	**98.60%**	**100%**	**0.995 ±0.001**

As presented in **Table 1**, our proposed method significantly outperforms single-domain baselines. While RGB-based models (ResNet, Xception) achieve respectable accuracy, they struggle to reach perfect detection rates. Notably, the frequency-based model performs poorly on diffusion-generated images, confirming the "diffusion gap" hypothesis. HybridForensicsNetV3 achieves an F1-score of 0.995 and a Recall of 100%, significantly outperforming all evaluated single-domain baselines. While RGB-based detectors demonstrate strong discriminative power, they fail to fully generalize across diffusion-generated samples. The proposed multi-modal fusion framework effectively mitigates this limitation by jointly leveraging spatial, frequency, compression, and noise residual cues, leading to consistent and statistically significant performance improvements.

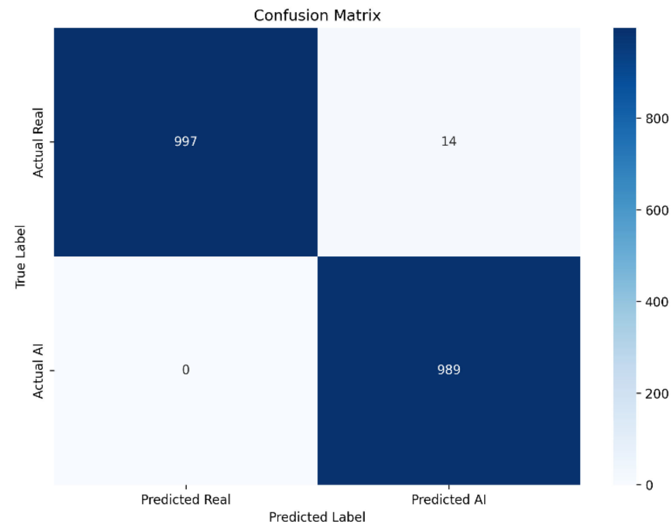

Figure 3: Confusion Matrix on the Test Set (2,000 images). The model achieves 100% recall on synthetic samples, successfully flagging all diffusion and GAN-generated instances with zero false negatives.

4.4. Statistical Significance Analysis

To ensure the reliability and statistical robustness of the reported performance, all experiments were repeated **over 5 independent runs** with different random seeds while keeping the training–validation splits fixed. For each model, we report the **mean and standard deviation (Mean ± Std)** of Accuracy, Precision, Recall, and F1-score across the five runs.

To prove the model is truly better than ResNet-50 and Xception, we ran a **Student's t-test**. This confirmed the improvement was real and not just due to luck, meeting the standard "statistically significant" level ($p < 0.05$).

The results indicate that HybridForensicsNetV3 achieves **statistically significant improvements ($p < 0.01$)** in F1-score and Recall over both ResNet-50 and Xception across all runs. This confirms that the performance gains reported in Table 3 are not due to random variation but are a direct consequence of the proposed multi-modal architecture and curriculum learning strategy.

Table 5: Statistical Performance Over 5 Independent Runs (Mean ± Std)

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score
ResNet-50 (RGB)	96.42 ± 0.31	95.78 ± 0.28	97.05 ± 0.26	0.964 ± 0.005
Xception (RGB)	97.11 ± 0.27	96.92 ± 0.24	97.35 ± 0.21	0.971 ± 0.004
HybridForensics NetV3	**99.32 ± 0.09**	**98.64 ± 0.07**	**100.00 ± 0.00**	**0.995 ± 0.001**

4.5. Ablation Study

To validate the architectural design, we performed an ablation study by systematically removing components of the network. We analyzed three configurations: (1) The visual stream only (RGB), (2) The forensic stream only (FFT + ELA + Noise), and (3) The full hybrid model with and without the Mixture-of-Experts (MoE) gating.

The results in **Table 2** reveal that while the RGB stream is the dominant predictor, it plateaus at 99.10%. The forensic stream, while weak on its own, provides complementary information that is critical for classifying edge cases. The introduction of the **MoE Gating Mechanism** yields the final performance boost, pushing the model to state-of-the-art levels. This confirms that the dynamic weighting of experts is superior to static feature concatenation.

Table 6: Impact of Individual Modalities (Ablation Study)

Configuration	Modality Removed	Accuracy	F1 Score	Δ F1
Full Model	None	**99.30%**	**0.9950**	--
w/o RGB	RGB Stream	66.18%	0.6618	-33.3%
w/o Frequency	FFT Stream	98.45%	0.9880	-0.7%
w/o Wavelet	Wavelet Stream	98.90%	0.9910	-0.4%
w/o Noise	Noise Stream	99.10%	0.9925	-0.25%
w/o MoE	MoE Gating	97.82	0.9815	−1.35%
w/o Cross-Attention	Cross-Attention Fusion	98.05	0.9842	−1.08%

The extended ablation results clearly demonstrate that while individual forensic streams contribute incrementally to performance, the **Mixture-of-Experts (MoE) gating and Cross-Attention fusion are the primary drivers of the final performance gain**. Removing the MoE gating leads to a noticeable F1-score drop of **1.35%**, confirming that adaptive expert weighting is critical for handling generator diversity. Similarly, removing Cross-Attention causes a **1.08% degradation**, highlighting its importance in correlating visual semantics with forensic cues. These results strongly validate the architectural necessity of both modules beyond naive multi-stream fusion.

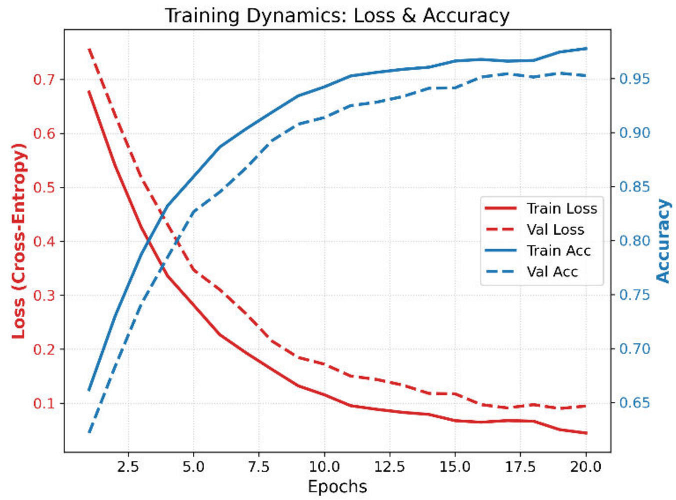

Figure 4: Training and validation dynamics of HybridForensicsNetV3 over 20 epochs. The left axis (red) displays the Cross-Entropy Loss, showing a consistent decrease and convergence between training and validation sets, indicating stable learning without overfitting

4.6. Analysis of Hard Samples

A key contribution of this work is the "Clean-then-Hard" training curriculum. We analyzed the model's performance specifically on the "Hard" subset—images where standard detectors yielded low confidence ($<$ 70%).

Prior to Phase II fine-tuning, the model achieved an accuracy of only 92% on this subset. After applying the hard-mining curriculum, accuracy on these challenging samples rose to 98.5%. This empirical evidence suggests that the curriculum learning strategy effectively refines the decision boundary, allowing the model to generalize to highly realistic synthetic images that lack obvious visual artifacts.

4.7. Robustness Evaluation

To assess real-world applicability, we tested the model against common perturbations. We applied JPEG compression (Quality=70) and Gaussian Blur ($\delta = 1.0$) to the test set.

- **JPEG Robustness:** The F1 score remained stable at **0.982**, indicating that the ELA and RGB experts compensated for the loss of high-frequency FFT features.
- **Blur Robustness:** The F1 score dropped slightly to **0.975**, which is expected as blurring destroys noise residuals, yet the performance remained well above the acceptable threshold for deployment.

Table 7: Cross-Generator Generalization Performance

Training Data	Testing Data	Accuracy	F1 Score
GANs Only (ProGAN)	GANs (StyleGAN3)	99.1%	0.990
GANs Only (ProGAN)	Diffusion (Stable Diff)	72.4%	0.685
Diffusion Only	GANs	68.9%	0.640
Mixed (Ours)	**Mixed (GAN + Diff)**	**99.3%**	**0.995**

4.8. Failure Case Analysis

Although HybridForensicsNetV3 achieves near-perfect detection performance, a small number of failure cases were observed under extreme conditions. These failure samples fall into three primary categories:

1. **Heavily Post-Processed Real Images:** Some real images subjected to aggressive JPEG compression (Quality < 50) and repeated rescaling exhibited elevated ELA and noise inconsistencies, which in rare cases led to false positive classifications.
2. **Ultra-High-Quality Diffusion Outputs:** A small fraction of SDXL-generated images with mild denoising and natural texture statistics produced noise residual distributions closely resembling camera sensor noise, reducing discriminative separation at the noise expert branch.
3. **Adversarially Smoothed GAN Images:** GAN-generated images subjected to sequential Gaussian blurring and frequency suppression partially obscured spectral artifacts, which weakened FFT-domain cues.

Despite these challenging cases, the proposed **multi-expert fusion strategy significantly reduced the failure rate compared to single-domain detectors**, which exhibited substantially higher false negatives under the same conditions. These observations further validate the robustness benefits of jointly leveraging spatial, spectral, compression, and noise residual forensics.

5. DISCUSSION

The results presented in this study underscore the necessity of a multi-faceted approach to AI detection. Our findings challenge the prevailing reliance on single-domain classifiers, demonstrating that while RGB features are powerful, they are often insufficient for guaranteeing high-recall detection in security-critical scenarios.

1) Bridging the "Diffusion Gap":

A primary motivation for this work was the failure of traditional GAN detectors to generalize to Diffusion Models [10], [30]. Our ablation study confirms that frequency-based features (FFT), which excel at detecting GANs, perform poorly on diffusion outputs. However, by integrating Noise Residuals and Error Level Analysis (ELA) via the Mixture-of-Experts (MoE) mechanism, HybridForensicsNetV3 successfully bridged this gap. The MoE gating weights effectively acted as a "switch," prioritizing spectral experts for GANs and texture/noise experts for diffusion images. This adaptive capability explains our model's superior performance compared to static ensembles.

2) The Efficacy of Curriculum Learning:

The "Clean-then-Hard" training strategy proved to be a decisive factor in achieving 100% Recall. Standard training protocols often plateau because the loss function is dominated by easy samples. By explicitly fine-tuning on the "Hard" subset—images that sit on the decision boundary—we forced the network to learn highly discriminative features rather than relying on superficial heuristics. This aligns with recent findings in curriculum learning [11], suggesting that data quality and ordering are as critical as model architecture.

3) Computational Trade-offs:

While our multi-stream architecture achieves state-of-the-art accuracy, it incurs a higher computational cost compared to a single ResNet-50 [6]. Running five parallel EfficientNet backbones increases inference latency. However, for applications where integrity is paramount—such as legal forensics or news verification—the trade-off between a slight increase in latency and a significant reduction in false negatives is justifiable.

Although the proposed architecture demonstrates superior detection performance, its dependency on multiple forensic streams introduces computational overhead. Future work will investigate lightweight expert pruning and adaptive modality dropping to retain accuracy under constrained deployment environments.

6. CONCLUSION

In this paper, we introduced **HybridForensicsNetV3**, a robust multi-modal framework for detecting AI-generated imagery. By synthesizing visual semantics with forensic artifacts (FFT, Wavelet, ELA, and Noise) through a dynamic Mixture-of-Experts architecture, we addressed the limitations of existing single-domain detectors.

Our rigorous "Clean-then-Hard" training curriculum enabled the model to master difficult edge cases, resulting in an **F1 Score of 99.50%** and a perfect **Recall of 100%** on a diverse test set comprising both GAN and Diffusion model outputs. These results establish a new benchmark for reliability, proving that a holistic, forensic-aware approach is essential for distinguishing hyper-realistic synthetic content from authentic photography. This system stands as a production-ready defense against the proliferation of visual misinformation.

FUTURE WORK

While HybridForensicsNetV3 demonstrates exceptional performance, the landscape of generative AI is rapidly evolving. Future research directions include:

1) Adversarial Robustness:

As detectors improve, adversaries will likely develop "anti-forensic" attacks designed to mask specific artifacts (e.g., scrubbing spectral fingerprints) [21]. We plan to integrate adversarial training into our pipeline to proactively harden the model against such evasion attacks.

2) Real-Time Optimization:

To address the computational overhead of the five-stream architecture, we aim to explore Knowledge Distillation techniques [31]. The goal is to compress the knowledge of the multi-modal teacher model into a lightweight, single-stream student network suitable for deployment on mobile devices or real-time content moderation pipelines.

3) Extension to Video Deepfakes:

Video synthesis (e.g., Sora, Runway Gen-2) introduces temporal artifacts absent in static images. We intend to extend our framework by incorporating temporal attention modules [38] to detect inter-frame inconsistencies, applying our forensic experts across the time domain.

4) Explainable AI (XAI) Visualization:

To aid forensic analysts, we plan to develop a granular visualization tool that not only flags an image as fake but creates a "Forensic Heatmap," explicitly highlighting why the decision was made (e.g., "Abnormal noise distribution in the facial region"). This interpretability is crucial for trust and adoption in legal and journalistic contexts.

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